Thin Solid Films 520 (2011) 563–568

Contents lists available at ScienceDirect

Thin Solid Films

journal homepage: www.elsevier.com/locate/tsf

Effects of high-temperature annealing on ultra-thin CdTe solar cells

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ARTICLE INFO

Article history:
Received 23 February 2011
Received in revised form 23 June 2011
Accepted 28 June 2011
Available online 7 July 2011

Keywords:
Cadmium telluride
Thin films
Annealing
Cadmium chloride
Surface treatment
Chemical treatment
Solar cells

ABSTRACT

High-temperature annealing (HTA), a process step prior to vapor cadmium chloride (VCC) treatment, has been found to be useful for improving the crystallinity of CdTe films and the efficiency of ultra-thin CdTe solar cells. Scanning electron microscopy, optical absorption, photoluminescence measurements and analyses on photoluminescence results using spectral deconvolution reveal that the additional HTA step produces substantial grain growth and reduces grain boundary defects. It also reduces excessive sulfur diffusion across the junction that can occur during the VCC treatment. The HTA step helps to produce pinhole-free CdTe films and reduce electrical shorts in ultra-thin CdTe solar cells. An efficiency of about 11.6% has been demonstrated for ultra-thin CdS/CdTe solar cells processed with HTA step.

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1. Introduction

Thin-film CdS/CdTe solar cells are based on crystalline CdTe films produced by vapor deposition. A common deposition method is close-space sublimation (CSS), where the source and substrate are held at close proximity and their temperature differential is kept relatively small (<50 °C) in order to promote crystalline CdTe film growth near equilibrium conditions at high temperature (~550 °C). The CSS-deposited film thickness is typically in the range of 4 to 8 μm [1–3], which is much more than necessary for complete light absorption in solar cells. Although thinner CdTe films are desirable, they are not commonly used because it is difficult to produce high-quality crystalline thin films that are also free of pinholes [4,5]. Comparing to the CSS method, other thin-film deposition methods [6,7] such as sputtering are better suited for the deposition of thin films. However, these alternative methods are not widely used for CdTe films because of their low deposition rates and other limitations [8]. From the industry standpoint, reducing CdTe usage in the solar cells would not only lower material cost and extend the tellurium supply, which is a limited reserve [6,9], but also ease the environmental problem of disposing a potentially hazardous material [6].

Apart from absorption issues, CdS/CdTe solar cells based on ultra-thin CdTe films (≤1.5 μm) are generally lower in efficiency as a result of increased grain boundaries and bulk defects in these films, which are necessarily of reduced grain size. It has been shown that excessive sulfur diffusion from CdS to CdTe can occur via the grain boundaries, causing degradation in the p–n junction [10,11]. Metal migration from the back contact to the front junction, also via the CdTe grain boundaries, is known to cause cell degradation [12]. Cadmium chloride treatment (VCC) is a standard process for fabricating thin film CdS/CdTe solar cells, known for improving CdTe crystallinity and reducing the lattice mismatch between CdS and CdTe. Generally it enhances the p-type characteristics of CdTe [13]. In this study, we have investigated the effects of VCC treatment and an additional high-temperature annealing (HTA) on ultra-thin, CSS-deposited CdTe films. We will show that HTA treatment prior to VCC treatment can significantly increase the crystalline quality of ultra-thin CdTe films without introducing pinhole defects and thereby improve the solar cell efficiency.

2. Experimental details

CdS films of this work were deposited on fluorine-doped tin oxide glass substrates (Pilkington, TEC-15) using a chemical bath deposition method [14]. They were then subjected to VCC treatment for 3–5 min, which was conducted in a CSS system with ultra-dry $CdCl_2$ granules as the source. The substrate and source temperatures were maintained at 390–410 °C in an O_2:N_2 (1:4) ambient of 6.67 kPa.

CdTe films were deposited on the CdS-coated substrates by CSS method [15] in oxygen ambient. The CdTe source was CdTe granules (Alfa Aesar, 99.99%) contained in a graphite holder. The gap between the CdTe source and the glass substrate was 4.5 mm. The deposition chamber ambience was typically 200 Pa oxygen. A typical temperature-time profile for CdTe deposition is shown in Fig. 1. The thickness of the CdTe film produced using this profile is approximately 1.5 μm.

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0040-6090/$ – see front matter. Published by Elsevier B.V.
doi:10.1016/j.tsf.2011.06.097

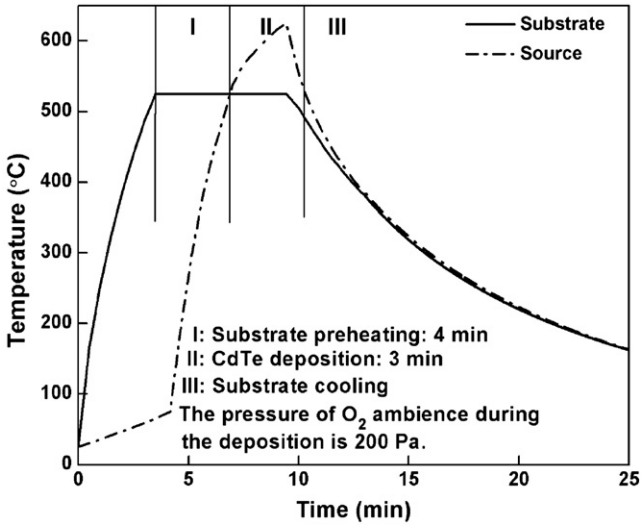

Fig. 1. A typical temperature profile for the deposition of ultra-thin CdTe films using the CSS method.

The HTA treatment was carried out in a tube furnace under nitrogen at a temperature up to 560 °C for a duration up to 45 min. Nitrogen flow under atmospheric pressure was maintained during the entire process.

Following HTA treatment, the CdS/CdTe coated substrates were subjected to another VCC treatment for 2 to 6 min. The substrate and source temperatures were maintained at 390–410 °C.

The back contact was MoO_x/Ni comprising a 40-nm MoO_x as the buffer layer [16] deposited by thermal evaporation on CdTe surface and a 200-nm Ni layer as the electrode deposited by DC sputtering. Prior to forming the back contact, the CdTe film was rinsed with

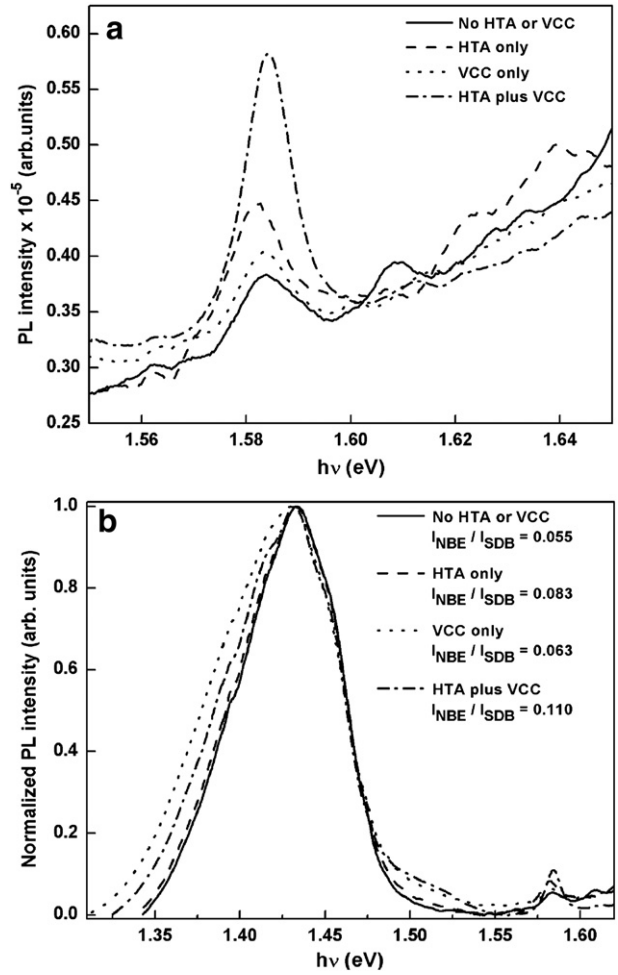

Fig. 3. PL spectra with surface excitation for: CSS-deposited ultra-thin CdTe film without any treatment, with VCC treatment, with HTA treatment, and with HTA + VCC treatments.

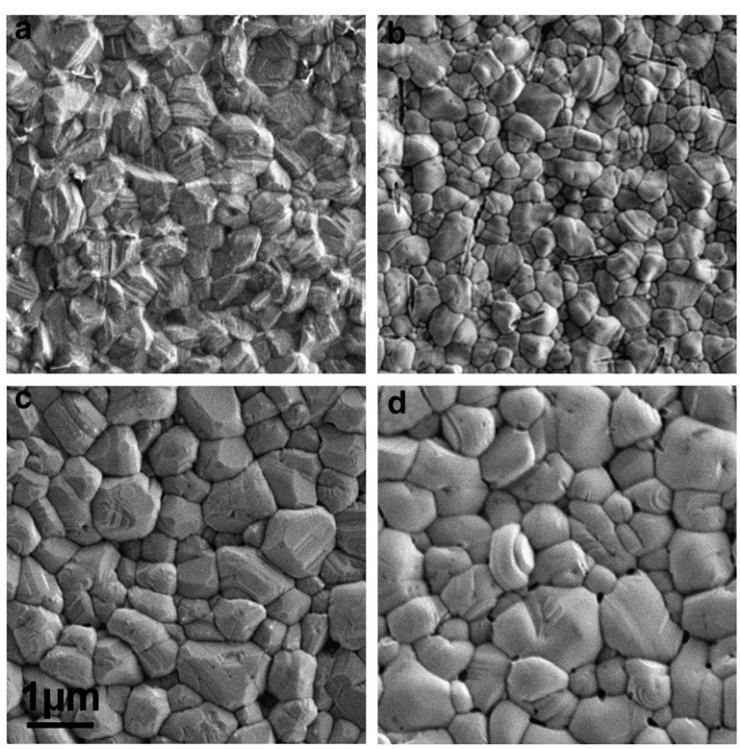

Fig. 2. SEM images for: CSS-deposited ultra-thin CdTe film (a) without any treatment, (b) with VCC treatment, (c) with HTA treatment, and (d) with HTA + VCC treatments.

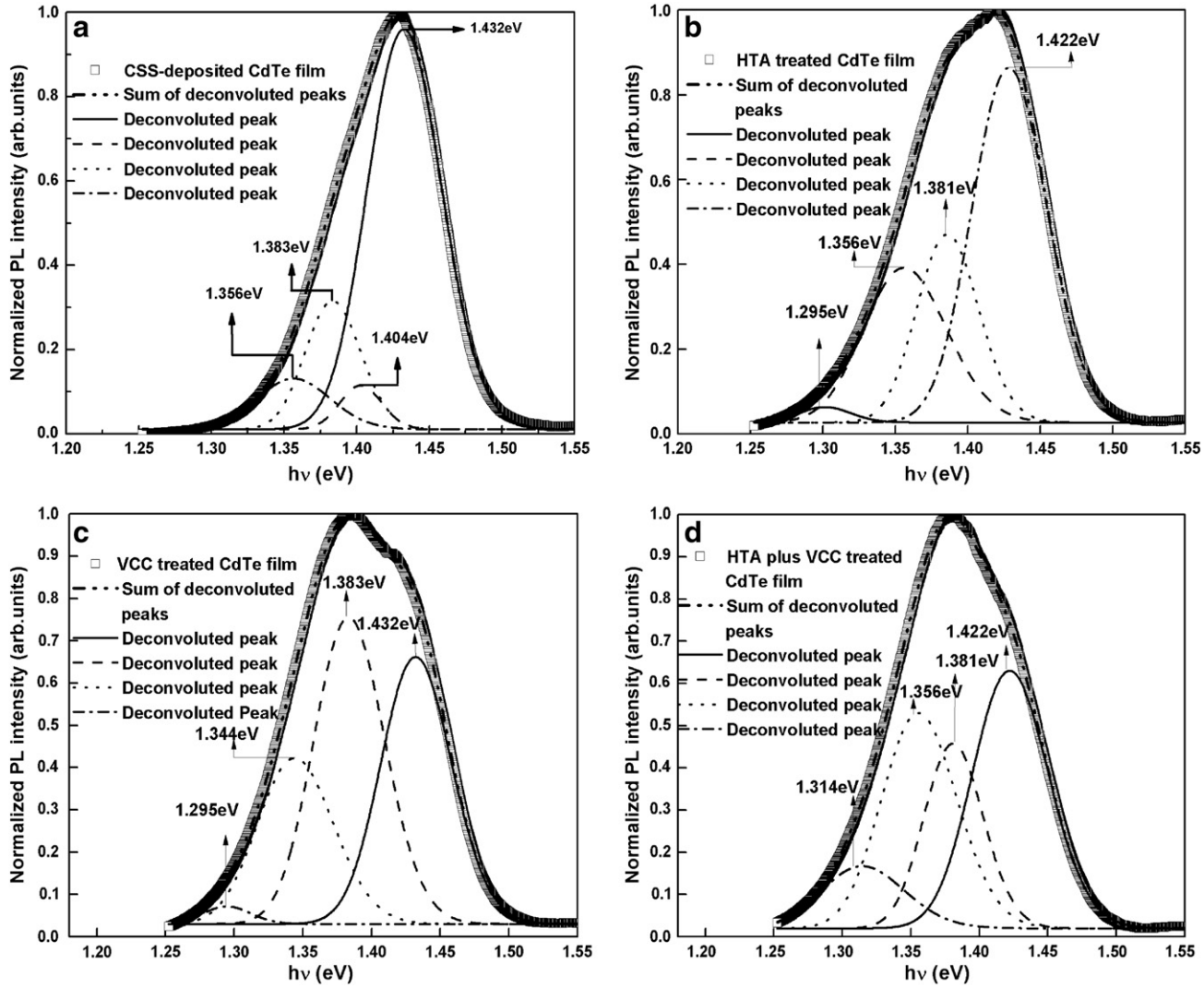

Fig. 4. PL spectra and the corresponding deconvoluted PL peaks for CSS-deposited ultra-thin CdTe film (a) without any treatment, (b) with HTA treatment, (c) with VCC treatment, and (d) with HTA + VCC treatments. PL spectra were obtained with interface excitation and at 40 K.

deionized water to remove any $CdCl_2$ residue from the VCC treatment process. The active area of the CdTe solar cells was 0.10 ± 0.005 cm^2, which was defined by mechanically scribing the CdS/CdTe/Ni layer stack with a razor blade.

The morphology of the CdTe films was examined using scanning electron microscopy (SEM) (Zeiss, Supra 40VP, operating voltage: 20 kV). The thickness of the CdTe films was measured using a surface profilometer (KLA-Tencor Alpha Step D-120). Optical absorption measurements were obtained with a Perkin-Elmer Lambda 900 spectrophotometer. Photoluminescence (PL) measurements were carried out using a He–Ne laser (Research Electro-Optical, 30992, 12 mW, 633 nm) as the excitation source and fiber optic spectrometer (Ocean Optics, USB 4000) as the detector. The solar cell current–voltage (*J*–*V*) data were obtained using a Keithley source meter (Model 2400) and a tungsten-lamp-based solar simulator (Solux 3SS4736-50 W solar simulator). The solar simulator was calibrated with a silicon photodiode (Hamamatsu S1787-12). The incident light intensity on the solar cell was 80 mW/cm^2. The values of the current density were corrected by integrating the photo-response of the cells with AM 1.5 solar spectrum. Spectral response measurements were obtained using a calibrated 1/4 m monochromator (ARC SpectroPro 275).

3. Results and discussions

In Fig. 2 the SEM images (a–d) show the evolution of CdTe grain growth as a result of HTA and VCC treatments. The CSS-deposited CdTe film without any treatment (Image a) has a relatively small grain size of about 0.5 μm. VCC treatment on the CdTe film (Image b) does not appear to change the grain size significantly, although it tends to smooth the grain facets. On the other hand, HTA treatment induces large CdTe grain growth as shown in Image c. The average grain size of the HTA-treated film is significantly larger than that of the untreated film. As shown in Image d, subsequent VCC treatment of the HTA-treated film produces further grain growth as well as more rounded grain facets.

Fig. 3 shows the PL spectra of these CdTe films with excitation directed at the free CdTe surface. The prominent features are bands with peaks at ~1.430 eV and ~1.585 eV. The ~1.585 eV band, known as the "near-band-edge (NBE) band", has been attributed to bound excitons [17,18] and used as an indicator of CdTe crystallinity [19]. As shown in Fig. 3a, the NBE peak increases significantly with HTA and VCC treatments, indicating improvements in CdTe crystallinity. Known as the "surface defect band (SDB)", the ~1.430 eV band is presumably due to intrinsic defects [17]. As shown in Fig. 3b, the NBE/SDB intensity ratio also increases with HTA and VCC treatments,

further supporting that the CdTe quality is improved. The NBE peak intensity and NBE/SDB intensity ratio are the highest for the CdTe film with HTA + VCC treatment, which corresponds to the largest CdTe grains as shown in Fig. 2.

The CdS/CdTe interface was also characterized by PL measurements. The PL spectra, taken with excitation through the substrate, feature a broad peak around 1.4 eV, which broadens and shifts to lower energy with HTA and VCC treatments. As Fig. 4 shows, to analyze the evolution of these spectra, they were deconvoluted into four Gaussian components using *Origin Pro 8.0* with peaks located at ~1.43, ~1.38, ~1.35, and ~1.30 eV, which have been attributed to various species in the literatures. The ~1.43 eV band is the surface defect band also observed from the free surface. The ~1.38 eV band has been assigned to $CdTe_{1-x}S_x$ [20] and the ~1.35 eV band to cadmium vacancy complexes [17]. The small ~1.30 eV band, which is only observed in CdTe with treatments, has not been identified. Fig. 5a shows that the intensity of the defect band is reduced with HTA and VCC treatments, suggesting that the defect concentration at the CdS/CdTe interface is reduced [21]. It is known that VCC treatment can promote sulfur diffusion from CdS to CdTe [22,23]. As Fig. 5b shows, the intensity of the ~1.38 eV band increases significantly with VCC treatment, a clear evidence that $CdTe_{1-x}S_x$ is being formed near the interface. It is also noteworthy that the ~1.38 eV band for the films with HTA and HTA + VCC treatments is considerably weaker compared to the film with VCC treatment only, suggesting the additional HTA treatment can mitigate excessive sulfur diffusion in the VCC treatment.

Fig. 6a shows the absorption of ultra-thin CdTe films near the absorption edge region. It is evident that the absorption edge shifts to lower energy with post-deposition treatments, especially the VCC treatment. This red shift is consistent with the formation of $CdTe_{1-x}S_x$, a low bandgap component resulting from the incorporation of sulfur in CdTe films with these treatments. The bandgap of $CdTe_{1-x}S_x$ was extracted from the absorption spectra using a linear fit at the threshold region [23], as illustrated in Fig. 6a. Using the extracted bandgap of $CdTe_{1-x}S_x$ and an established relation between the bandgap and stoichiometry of $CdTe_{1-x}S_x$ [24,25]:

$$E_g(x) = 1.74 \cdot x^2 - 1.01 \cdot x + 1.51 \quad (1)$$

the values of x are calculated and presented in Fig. 6b with relation to the bandgap of $CdTe_{1-x}S_x$.

HTA itself can promote the sulfur diffusion from CdS to CdTe as a result of high temperature, causing the x value in $CdTe_{1-x}S_x$ to increase from 0.8% in the cell without any treatment to 1.3% in the cell with just HTA treatment. However, HTA is less effective compared to VCC treatment in promoting sulfur diffusion with the latter producing an apparently higher x value of 2.3% in the cell with just VCC treatment despite shorter treatment duration. Furthermore, it is noteworthy that HTA treatment prior to VCC treatment can suppress

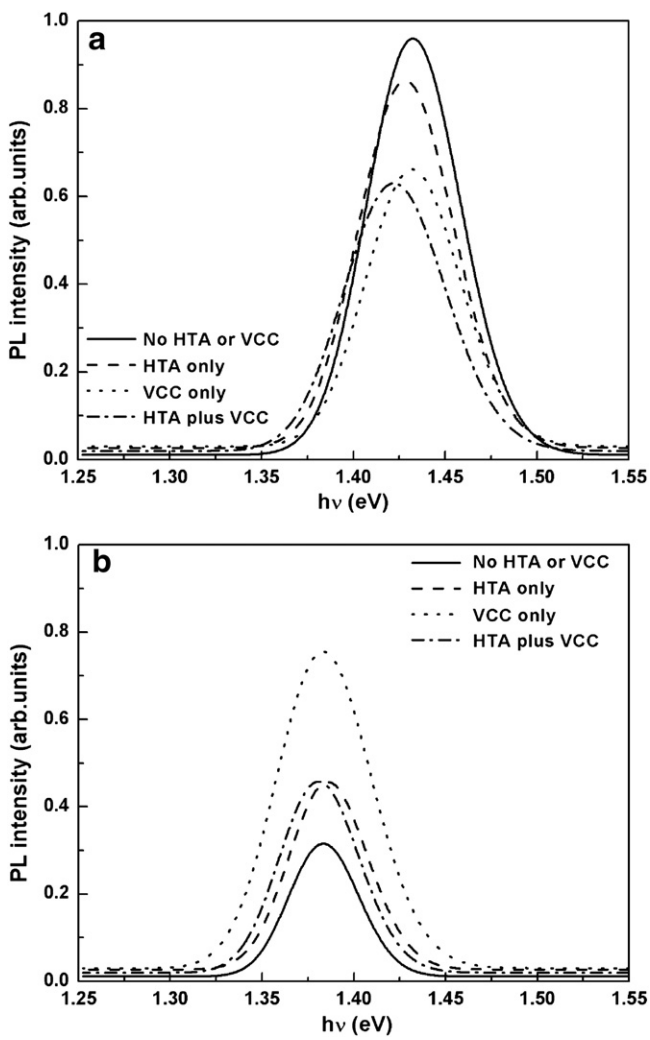

Fig. 5. PL spectra with interface excitation showing the evolution of: (a) the interface defect band, and (b) the $CdTe_{1-x}S_x$ band with various HTA and VCC treatments.

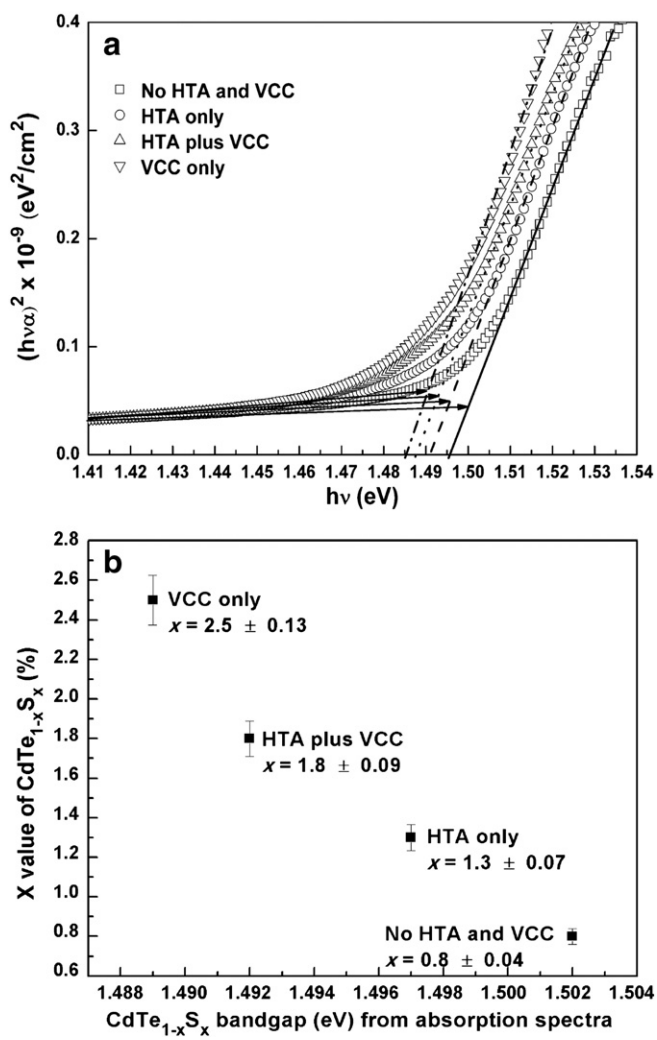

Fig. 6. (a) Absorption at the band edge region for ultra-thin CdTe films with various HTA and VCC treatments. (b) the x values of $CdTe_{1-x}S_x$ in relation to the bandgap of $CdTe_{1-x}S_x$.

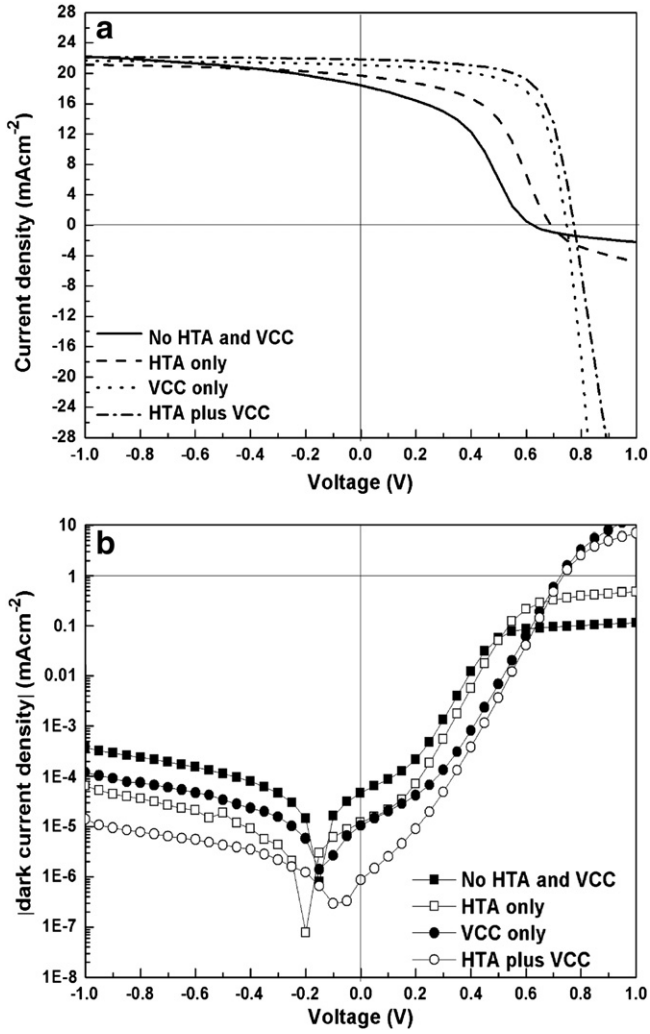

Fig. 7. (a) Photo and (b) dark *J–V* curves for ultra-thin CdS/CdTe solar cells with various HTA and VCC treatments.

sulfur diffusion resulting from the VCC treatment. This can be seen in the cell with HTA plus VCC treatment where *x* value is only 1.8%. Sulfur diffusion from CdS to CdTe can proceed via Fickian bulk and

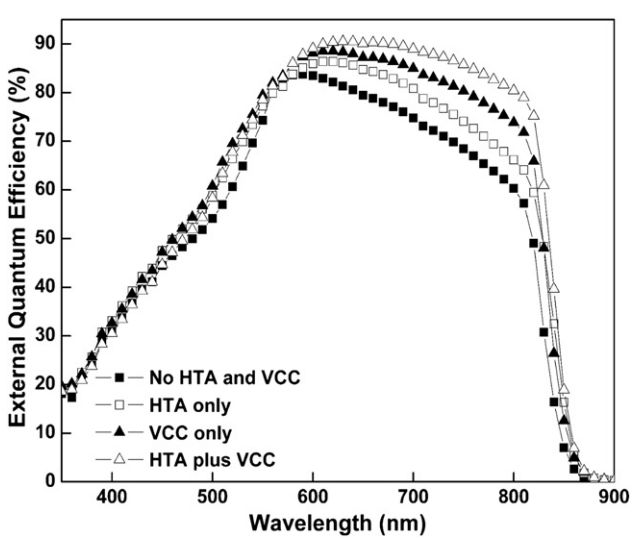

Fig. 8. Spectral response curves for ultra-thin CdS/CdTe solar cells with various HTA and VCC treatments.

Table 1
Device performances of ultra-thin CdS/CdTe solar cells with various HTA and VCC treatments. The thickness of CdTe films is 1.5 μm.

Cell	HTA	VCC	Device performances					
			R_{sh} (Ω)	R_s (Ω)	J_{sc} (mA/cm^2)	V_{oc} (mV)	FF (%)	η (%)
1168A	N	N	1363	770	18.4	630	42.4	4.9
1168B	Y	N	3078	257	19.6	690	52.0	7.0
1168C	N	Y	8681	40	21.1	750	67.2	10.6
1168D	Y	Y	14,971	52	21.9	778	68.2	11.6

grain-boundary processes in the CdTe layer. The reported activation energy for these two processes are ~3.0 and ~2.0 eV, respectively [26], suggesting that sulfur diffusion via the grain boundaries is dynamically faster than that via the bulk. As inferred from the PL and absorption analysis, the suppression of sulfur diffusion by the HTA treatment is consistent with SEM observation (Fig. 2) showing large grain growth induced by HTA, which implies a reduction in grain boundaries in CdTe.

Fig. 7a and b shows the photo and dark *J–V* characteristics of ultra-thin CdTe solar cells with various treatments using MoO_x/Ni as the back contact. Compared to the reference cell (without any treatment), the cell with only HTA treatment exhibits significantly improvement in open-circuit voltage (V_{oc}), short-circuit current (J_{sc}) and fill factor (FF), although it is still substantially inferior to the cell with only VCC treatment. Combining HTA and VCC treatments further enhances the cell performance over the VCC treatment alone, as shown in Table 1. The improvement in V_{oc} is consistent with the reduction of reverse saturation current shown in Fig. 7b, indicating an increase in CdTe grain size due to HTA treatment does not necessarily compromise the shunt resistance in ultra-thin CdTe solar cells. Table 1 includes the dynamic shunt resistance, R_{sh}, and dynamic series resistance, R_s, calculated from the slope of d*J*/d*V* at J_{sc} and V_{oc}, respectively [27]. The cell with both HTA and VCC treatments yields excellent R_{sh} and R_s, as well as the highest efficiency of 11.6%.

Fig. 8 shows the spectral response curves for the ultra-thin CdTe cells with various HTA and VCC treatments. It is clear that the improvement in quantum efficiency is significantly higher in the long wavelength region and for cells that have been subjected to HTA and VCC treatments. The low quantum efficiency in ultra-thin CdTe solar cell, particularly near band edge region, can be attributed to incomplete light absorption [6,9] and recombination loss at the back contact [6]. According to the analysis of the absorption edge in Fig. 6, the red shift in the band edge region is due to $CdTe_{1-x}S_x$, the formation of which is enhanced with HTA and VCC treatments. Thus, the increase in quantum efficiency can be partly due to increased $CdTe_{1-x}S_x$ absorption. Reduction in the recombination loss at the back contact can also contribute to the increase in quantum efficiency. This implies improved crystal quality in the ultra-thin CdTe films, which are supported by PL, absorption, and SEM characterizations.

We have performed preliminary light soaking tests on the ultra-thin CdTe cells. Under an equivalent of approximately one sun illumination (compact fluorescent light source and 1 kW/m^2) and laboratory environment and temperature of 45 °C, cells with and without HTA treatments decay approximately 4–5% after 1000 h of continuous illumination.

4. Conclusions

We have found high-temperature annealing (HTA), an additional process step prior to vapor cadmium chloride (VCC) treatment, to be useful for producing high-quality ultra-thin CdTe films. SEM, optical absorption and PL measurements reveal substantial growth in CdTe grain size and reduction in defects related to grain boundary and excessive sulfur diffusion. An efficiency of about 11.6% has been

demonstrated for ultra-thin CdS/CdTe solar cells processed with the HTA step.

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